TWO-ROAD TRAFFIC-LIGHT CONTROLLER

Moore Finite State Machine (FSM) — Design Report

Purpose: Control two intersecting roads so that only one road receives a green signal at a time. Yellow signals provide transition warning, and an all-red clearance state separates the two directions.

1. State assignment and output table

State	Q2 Q1 Q0	Road A	Road B	Next state
S0	000	Green	Red	S1
S1	001	Yellow	Red	S2
S2	010	Red	Red	S3
S3	011	Red	Green	S4
S4	100	Red	Yellow	S5
S5	101	Red	Red	S0
Unused	110, 111	Red	Red	S0 (recovery)

2. State transition sequence

S0 → S1 → S2 → S3 → S4 → S5 → S0. Three D flip-flops (Q2, Q1, Q0) store the current state and share a common clock. Reset initializes the state register to 000 (S0).

3. Next-state equations

D2 = (Q0 · Q1 · NOT Q2) + (Q2 · NOT Q0 · NOT Q1)

D1 = NOT Q2 · (Q0 XOR Q1)

D0 = NOT Q0 · (NOT Q1 + NOT Q2)

These equations implement the six-state sequence and force unused states 110 and 111 to 000 on the next clock.

4. Moore output equations

A_GREEN = NOT Q2 · NOT Q1 · NOT Q0

A_YELLOW = NOT Q2 · NOT Q1 · Q0

A_RED = Q2 + Q1

B_GREEN = NOT Q2 · Q1 · Q0

B_YELLOW = Q2 · NOT Q1 · NOT Q0

B_RED = NOT (B_GREEN + B_YELLOW)

Outputs depend only on the current state, so this is a Moore FSM. Add a clock divider/timing circuit if each state must last a real-world interval; each clock edge advances one state.

5. K-map and implementation notes

Use Q2, Q1, Q0 as K-map variables. Treat unused codes 110 and 111 as recovery states with next state 000, not as don't-cares. The minimized equations above follow this defined transition table.

6. Companion Logisim file

The companion .circ file is a ROM-based combinational lookup block. Set STATE[2:0] to a state code. Output bits [8:6] are next-state bits D2,D1,D0; bits [5:0] are A_GREEN, A_YELLOW, A_RED, B_GREEN, B_YELLOW, B_RED. To make the complete sequential FSM, connect D2,D1,D0 to three D flip-flops, connect Q2,Q1,Q0 back to STATE[2:0], share a clock, and reset the register to 000.

Important: The .circ file supplies the next-state/output lookup logic, not the completed flip-flop feedback wiring. Open and verify it in your Logisim Evolution version, then add the three D flip-flops before treating it as a fully simulated sequential controller.